

Investigating positional accuracy by double integrating accelerometer data

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1 Introduction

Inertial Odometry has been studied with a variety of applications, whenever you need to know the precise location of something instantly, chances are that double integrating accelerometer data outpaces GPS solutions. From robotics to step trackers, they all rely on the property that the area under a accelerometer graph can provide enough data used to compute the displacement of the object. In this investigation we will explore the accuracy of double integrating accelerometer data recorded on a phone during travel to find path traveled to get to school and create a visual representation of the position with respect to time and comparing with GPS data.

2 Numerical Integration and code application

Recall that in kinematics integrating acceleration gives velocity, and integrating once again gives displacement. For discrete-time data, this cannot be solve analytically. Instead we refer to the trapezoidal method, illustrated below:

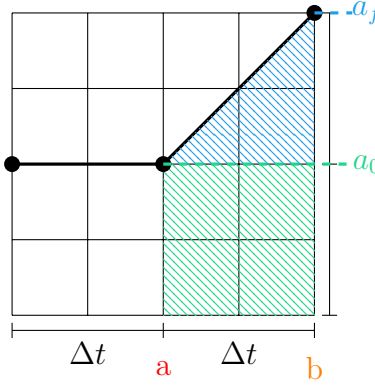


Figure 1: Integrating two data points

To calculate the area bounded by the x axis and the line created from two datapoints, we use basic geometry equations

$$\Delta v = \int_a^b a(t) dt = \Delta t \cdot a_0 + \frac{1}{2} \Delta t \cdot a_f \quad (1)$$

which is thus computed in python code

2.1 Testing the waters

the same idea will be repeated for each axis and afterwards applied to the calculated velocity values to find displacement. For this first model I moved my phone left and right 5 times on my table starting and ending at 0 velocity.

And integrating the x-axis with the aforementioned algorithm we get this series of integrations expressed graphically (Assume this part will be corrected)

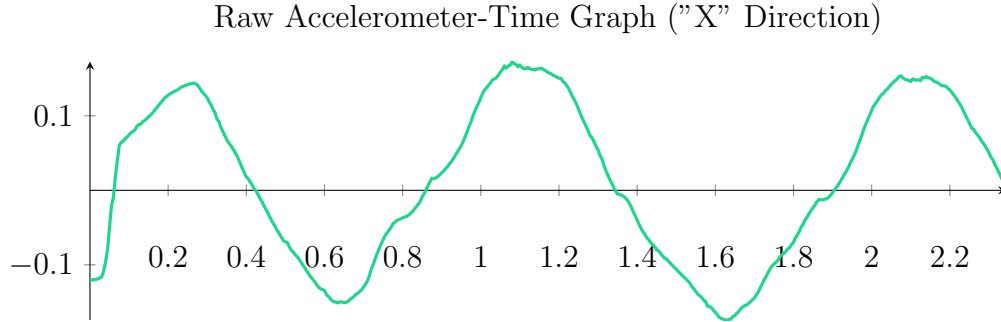


Figure 2: Raw accelerometer data on the X axis

Plugging into desmos for Sinusoidal regression we get:

$$y = 0.14272 \cdot \sin(6.6486x + 0.216978) - 0.00389886 \quad (2)$$

seen in figure 4. We will integrate this analytically twice:

$$v(t) = \int (a \cdot \sin(bt + c) - d) dt \quad (3)$$

where a, b, c, and d are constants. Rearranged, we get

$$v(t) = a \int \sin(bt + c) dt - \int d dt \quad (4)$$

Using u-substitution

$$u = bt + c \quad (5)$$

$$\frac{du}{dt} = b \quad (6)$$

$$dt = \frac{du}{b} \quad (7)$$

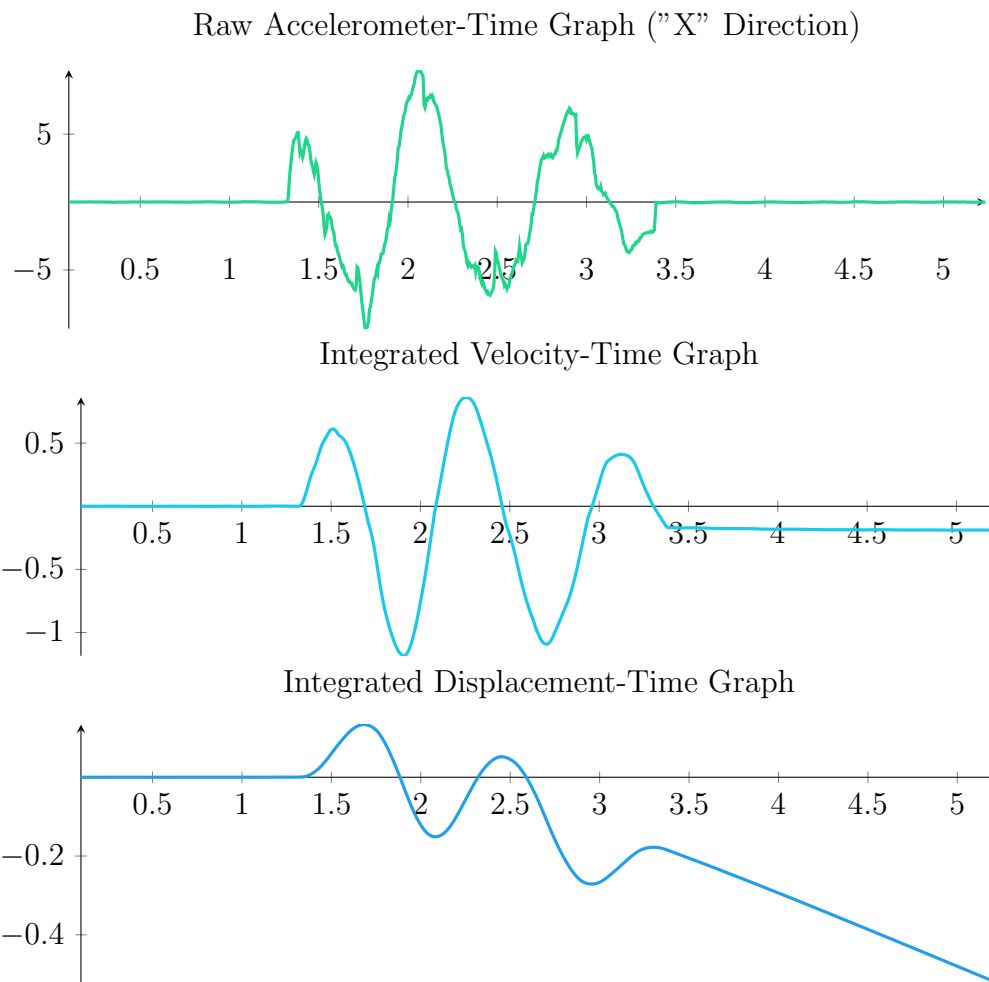


Figure 3: The Progressive Progress of Numerically integrating accelerometer data

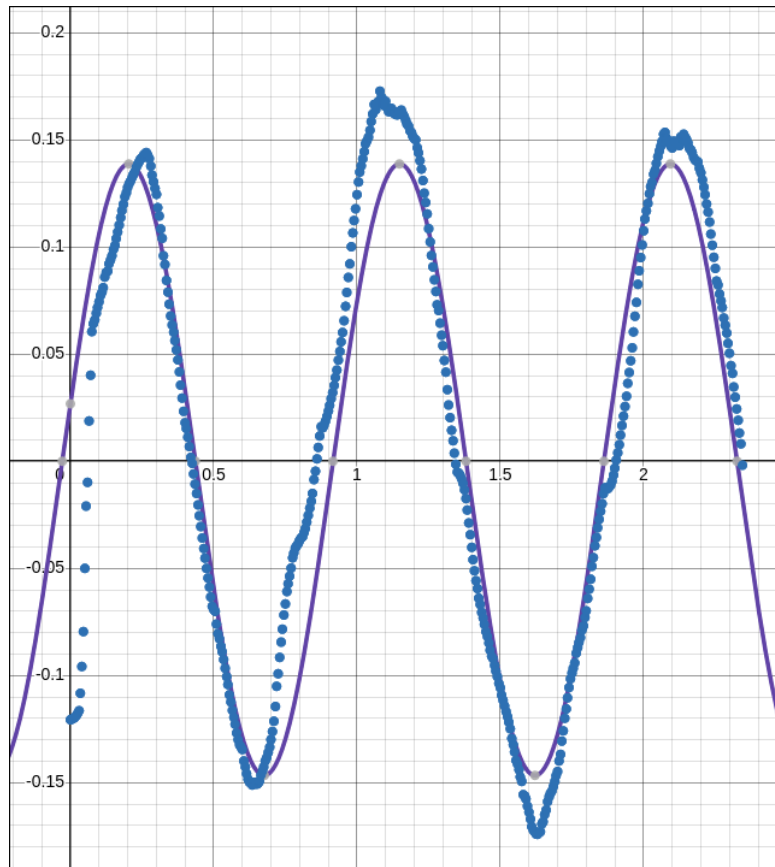


Figure 4: graph of our accelerometer data and the fitted curve

plugging in u and dt :

$$v(t) = a \int \sin(bt + c) dt - \int d dt \quad (8)$$

$$= a \int \sin(u) \frac{1}{b} du - \int d dt \quad (9)$$

$$= \frac{a}{b} \int \sin(u) du - \int d dt \quad (10)$$

$$= -\frac{a}{b} \cos(u) - \int d dt \quad (11)$$

$$= -\frac{a}{b} \cos(bt + c) - dt + C_1 \quad (12)$$

and since we know the constant represents the initial velocity, and since we start at rest, we need to find the constant C_1 such that $v(0) = 0$

$$v(0) = -\frac{a}{b} \cos((0)b + c) - (0)d + C_1 = 0 \quad (13)$$

$$\frac{a}{b} \cos(c) = C_1 = 0.0209628448456 \quad (14)$$

$$(15)$$

For now, we will keep in mind that $C_1 = 0.0209628448456$ but continue using C_1 throughout integration for convenience. To find the displacement function with respect to time we integrate again.

$$s(t) = \int \left(-\frac{a}{b} \cos(bt + c) - dt + C_1 \right) dt \quad (16)$$

$$s(t) = -\frac{a}{b} \int \cos(bt + c) dt - \int dt dt + \int C_1 dt \quad (17)$$

$$s(t) = -\frac{a}{b} \int \cos(bt + c) dt - \int dt dt + \int C_1 dt \quad (18)$$

$$(19)$$

Using the same u-substitution:

$$s(t) = -\frac{a}{b} \int \cos(u) \frac{du}{b} - \int dt \, dt + \int C_1 \, dt \quad (20)$$

$$s(t) = -\frac{a}{b^2} \sin(u) - \int dt \, dt + \int C_1 \, dt + C_2 \quad (21)$$

$$s(t) = -\frac{a}{b^2} \sin(bt + c) - \int dt \, dt + \int C_1 \, dt + C_2 \quad (22)$$

$$s(t) = -\frac{a}{b^2} \sin(bt + c) - \frac{dt^2}{2} + C_1 t + C_2 \quad (23)$$

$$(24)$$

and since this data is filmed as relative displacement from original point (seen as 0) we need to find the value of C_2 such that $s(0) = 0$. [Ai said wording was bad, I don't understand why]

$$s(0) = -\frac{a}{b^2} \sin(b(0) + c) - \frac{d(0)^2}{2} + C_1(0) + C_2 = 0 \quad (25)$$

$$0 = -\frac{a}{b^2} \sin(b(0) + c) + C_2 \quad (26)$$

$$\frac{a}{b^2} \sin(b(0) + c) = C_2 = 0.000695067592524 \quad (27)$$

$$(28)$$

Again we will keep $C_2 = 0.000695067592524$ in mind. Resulting in a final function $s(t)$:

$$s(t) = -\frac{a}{b^2} \sin(bt + c) - \frac{dt^2}{2} + C_1 t + C_2 \quad (29)$$

drawing a graph of progressive integration is seen in figure 5

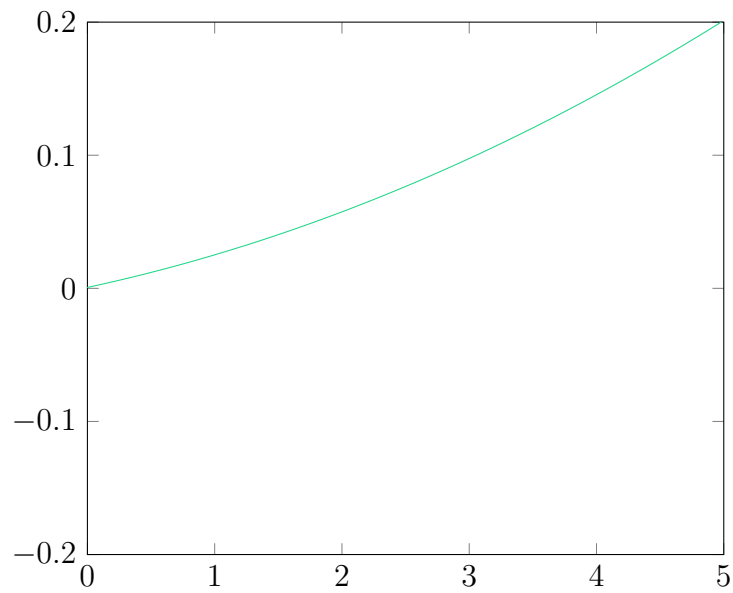


Figure 5: graph of the model being integrated twice

???

We have a displacement/time graph from video. However it is a different sampling rate (120hz) than the recorded datapoints (100hz). Hence we will attempt to align by drawing a line between the points in the recorded accelerometer data, and interpolate the value.

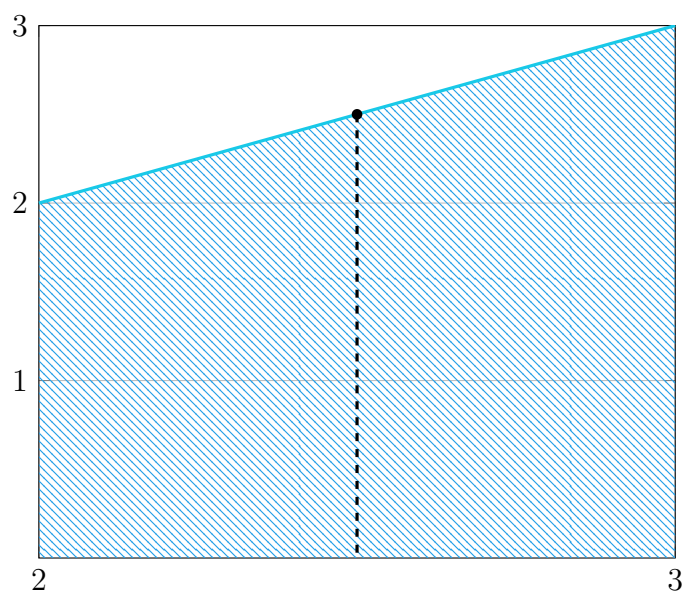


Figure 6: Illustration of interpolating values from two datapoints $x = 2, 3$